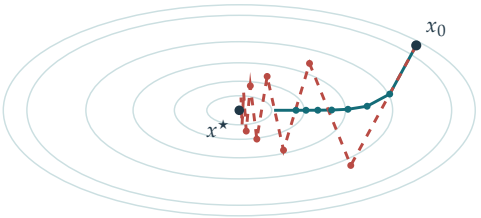
Panel 1

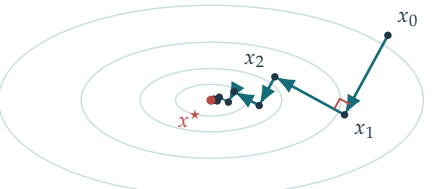
$f(x) = \frac{1}{2}(x_1^2 + 5x_2^2)$



A larger step can oscillate
across the narrow direction

Panel 2

$\tau_k \in \underset{\tau \geq 0}{\operatorname{argmin}}\, f(x_k - \tau \nabla f(x_k))$



$\langle \nabla f(x_k), \nabla f(x_{k+1}) \rangle = 0$

$f(x) = \frac{1}{2}(x_1^2 + 5x_2^2)$